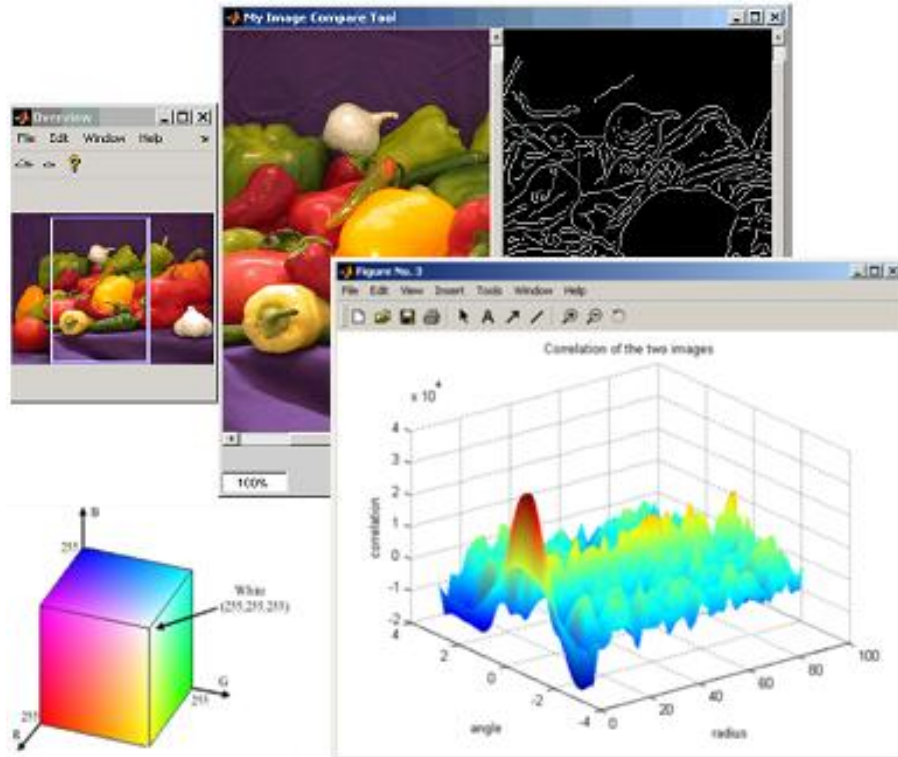


Digital Image Processing

LECTURE -3: DIGITAL IMAGE SIZE DETERMINATION

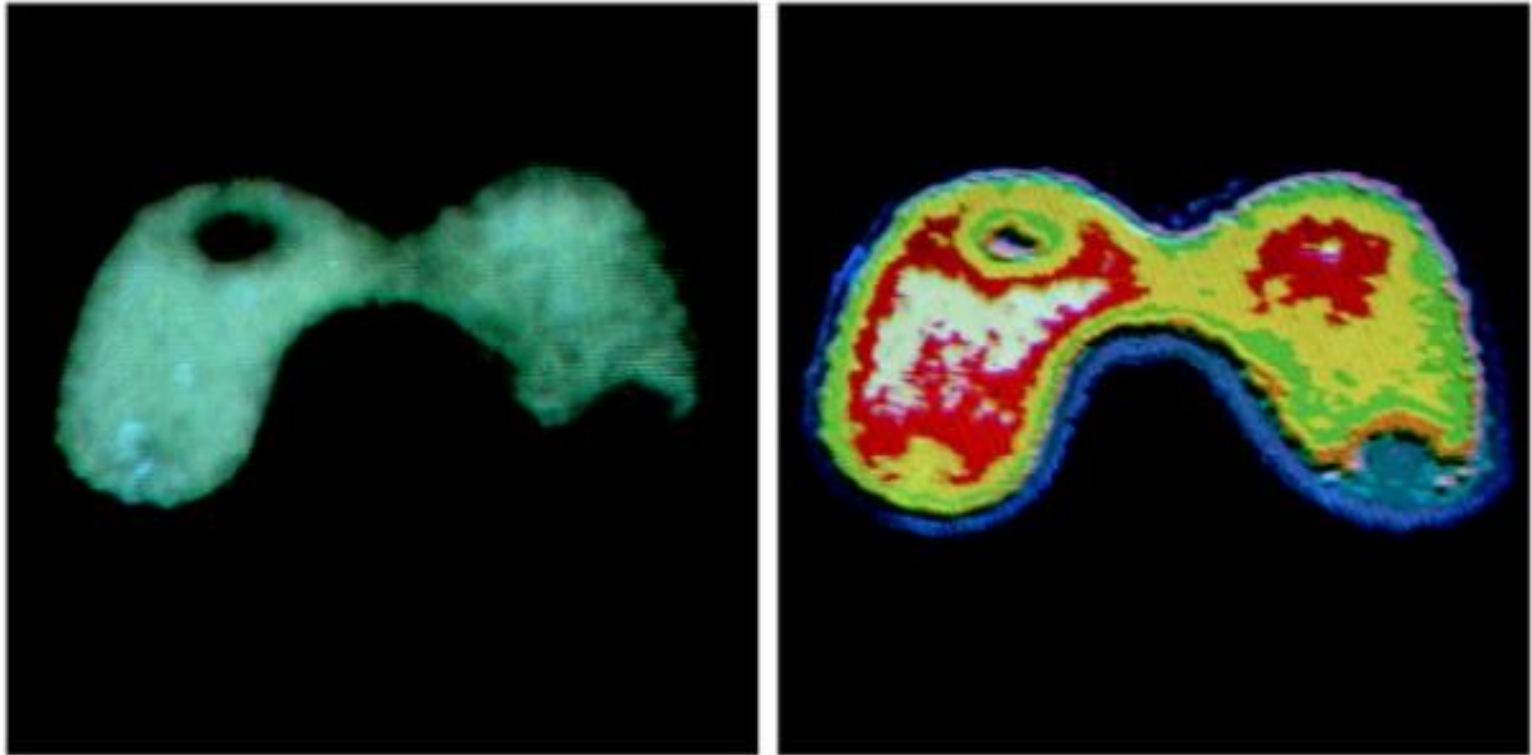


Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura

Motivation For Digital Image Processing

Motivation for Digital Image Processing

- It is motivated by two major applications:
 - Improvement of pictorial information for human perception



a b

FIGURE 6.20 (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

Motivation for Digital Image Processing

□ It is motivated by two major applications:

- Image processing for autonomous machine application



Motivation for Digital Image Processing

- It is motivated by two major applications:
 - Efficient storage and transmission

Original Image A



Reduced by 2



Reduced by 4



Reduced by 8



Applications of Digital Image Processing

Applications of Digital Image Processing

❑ Office Automation: Optical character recognition, document processing; cursive script recognition; logo and icon recognition, identification of address on envelope

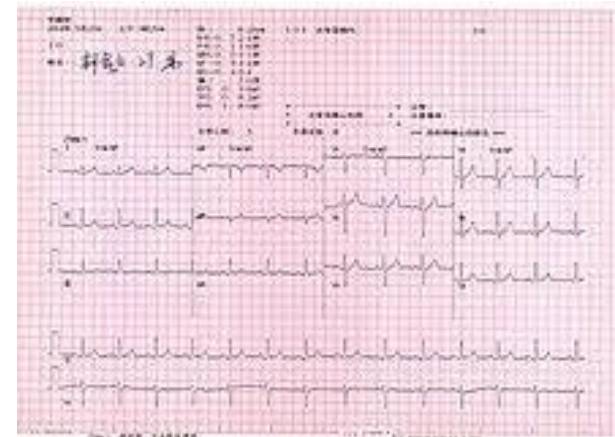


Cambridge

[illegible]

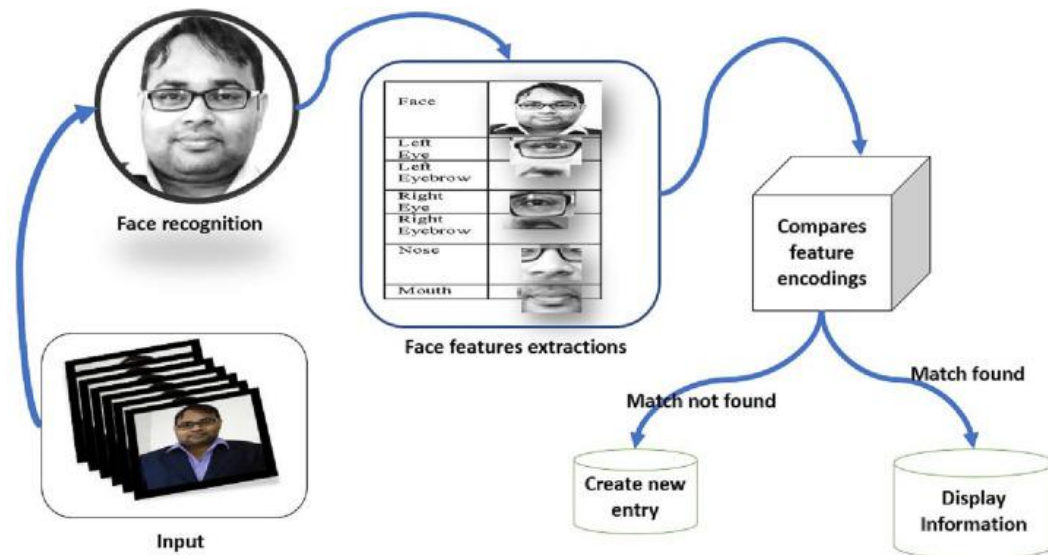
Applications of Digital Image Processing

❑ **Bio-medical:** ECG, EEG, EMG analysis, Cytological, historical and stereological applications, automated radiology and pathology, X-ray image analysis, mass screening of medical image



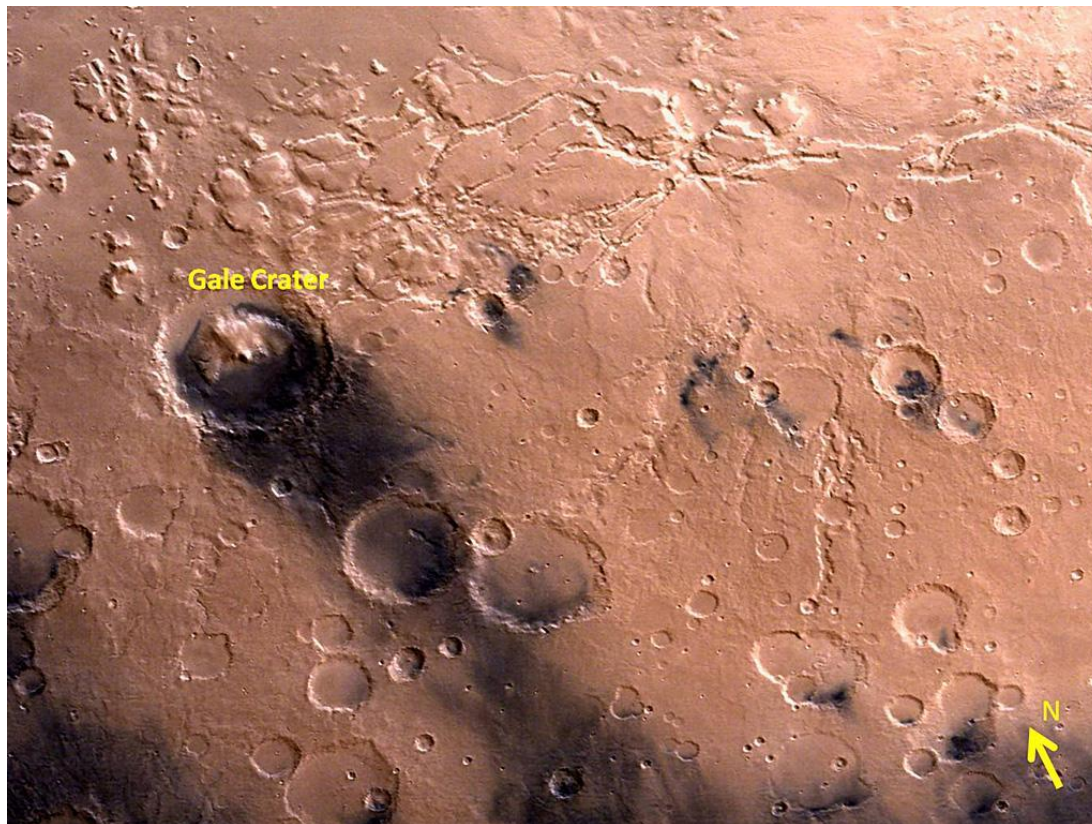
Applications of Digital Image Processing

☐ **Criminology:** Finger-print identification, human face registration and matching, forensic investigation.



Applications of Digital Image Processing

❑ **Astronomy and Space applications:** Restoration of image suffering from geometric and photometric distortions, computing close-up picture of planetary surfaces.



NB4 MgII h 280 nm 2024-May-17 05:20:26



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Applications of Digital Image Processing

☐ **Meteorology:** Short-term weather forecasting, long-term climatic change detection from satellite and other remote sensing data, cloud pattern analysis.



Applications of Digital Image Processing

- ❑ **Industrial Automation:** Automatic inspection systems; non-destructive testing, automatic assembling, process related to VLSI manufacturing, PCB checking; robotics, oil and natural gas exploration, seismography.
- ❑ **Remote Sensing:** Natural resources survey and management, estimation related to agriculture, hydrology, forestry, mineralogy, urban planning, environment and pollution control
- ❑ **Military Applications:** Missile guidance and detection, target identification, navigation of pilotless vehicle. Reconnaissance and range finding
- ❑ **Information Technology:** Facsimile image transmission, videotex, video-conferencing and videophones.
- ❑ **Entertainment and Consumer Electronics:** HDTV, multimedia and video-editing

Advantages of Digital Image Processing

- **Improved image quality:** Digital image processing algorithms can improve the visual quality of images, making them clearer, sharper, and more informative.
- **Automated image-based tasks:** Digital image processing can automate many image-based tasks, such as object recognition, pattern detection, and measurement.
- **Increased efficiency:** Digital image processing algorithms can process images much faster than humans, making it possible to analyze large amounts of data in a short amount of time.
- **Increased accuracy:** Digital image processing algorithms can provide more accurate results than humans, especially for tasks that require precise measurements or quantitative analysis.

Disadvantages of Digital Image Processing

- **High computational cost:** Some digital image processing algorithms are computationally intensive and require significant computational resources.
- **Dependence on quality of input:** Poor quality input images can result in poor quality output.
- **Limitations of algorithms:** Digital image processing algorithms have limitations, such as the difficulty of recognizing objects in cluttered or poorly lit scenes, or the inability to recognize objects with significant deformations or occlusions.
- **Dependence on good training data:** Poor quality training data can result in poor performance of the algorithm

Digital Image Size Determination

Digital Image Representation

- Monochrome image can be represented as a two-dimensional light intensity function $f(x, y)$
 - where
 - x & y are spatial coordinates
 - the value of f at point (x, y) is the gray level or the brightness of the image

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & f(0, 2) & \dots & \dots & f(0, Y - 1) \\ f(1, 0) & f(1, 1) & f(1, 2) & \dots & \dots & f(1, Y - 1) \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ f(X - 1, 0) & f(X - 1, 1) & f(X - 1, 2) & \dots & \dots & f(X - 1, Y - 1) \end{bmatrix}$$

Digital Image Representation

- Generally, image is of size $m \times n$
where,
 - m: number of rows and
 - n: number of columns
 - k: number of bits used for representation
- For an image of size $m \times n$, if k are the number of bits for representation, then

$$\text{Image size} = m \times n \times k$$

Numerical Example 1

Ques 1: For an image of 512 by 512 pixels, with 8 bits per pixel. Determine the image size?

Numerical Example 2

Ques 2: What is the storage requirement for a 1024×1024 binary image.

Numerical Example 3

Ques 3: If a 375×300 gray-scale image needs to be sent across the channel of capacity 28 kbps, then how much transmission time is required.

Assuming number of bits per pixel = 8

Numerical Example 4

Ques4: For an image of 64 by 64 pixels with 3 bands, with 4 bits per pixel, Determine the image size.

Numerical Example 5

Ques 5: For an image of 32 by 32 pixels with 4 bands, in which the smallest unit of an image consists of 11 shades, Determine the image size?

Numerical Example 6

Ques6: For an image of 25 by 30 pixels with 5 spectral, the pixels have 17 levels , Determine the image size?

Numerical Example 7

Ques7: An image scan center needs to store CAD images whose resolution is 1024×1024 , with 64 gray levels. A total of 100 layers are present.

- A. How much storage is required in GB.
- B. Suppose this image is transferred via medium having a transfer rate of 128 K bits/second in the form of packets. The size of a packet is 100 bits and each packet is segregated by a start and a stop flag of 14 bits each. Compute the time required to transfer the image (in hrs).

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Numerical Example 8

Ques 8: A common measure of transmission for digital data is the baud rate, defined as symbols (bits in our case) per second. As a minimum, transmission is accomplished in packets consisting of 14 start bits, a byte (8 bit) of information and 14 stop bits. Using these facts, answer the following:

- (i) How many seconds would it take to transmit a sequence of 200 images of size 1280 X 960 pixels with 256 intensity levels using a 3 M-baud (10^6 bits/sec) modem? (This is a representative medium speed for digital subscriber line residential medium.)
- (ii) What would be the time using a 30 G-baud (10^9 bits/sec) modem?

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Numerical Example 9

Ques 9:

- i. Suppose there is a multispectral image of size 100×100 . This image has 4 bands and each band uses 256 Gray levels. This image need to be transmitted at rate of 10K bit per second. Calculate the time required to transmit the image in seconds.
- ii. How many different 100×100 binary images can exist?

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Numerical Example 10

Ques 10: Suppose there is a multispectral image of a size 100×100 . This image has 4 bands and each band is using 256 Gray levels. Find the size of the image in bits.

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Numerical Example 11

Ques 11:

1B. What will be the size in bits of an RGB image using 3 bits for each channel and having 1024 rows and 512 columns.

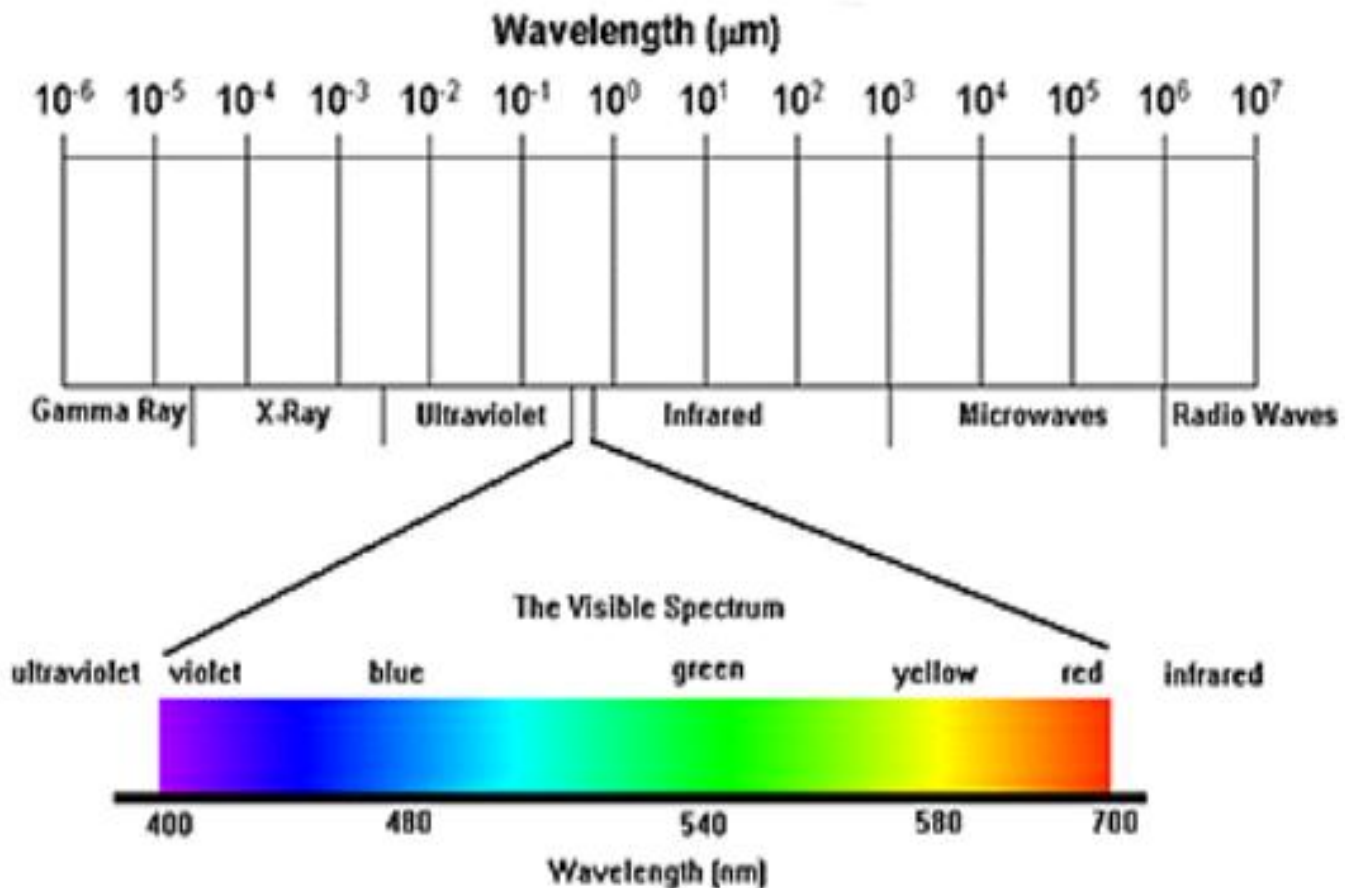
1C. How many different 10×10 binary images can exist.

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Image Sensing

Image Sensing

Sensing operates in various region of Electro Magnetic Spectrum

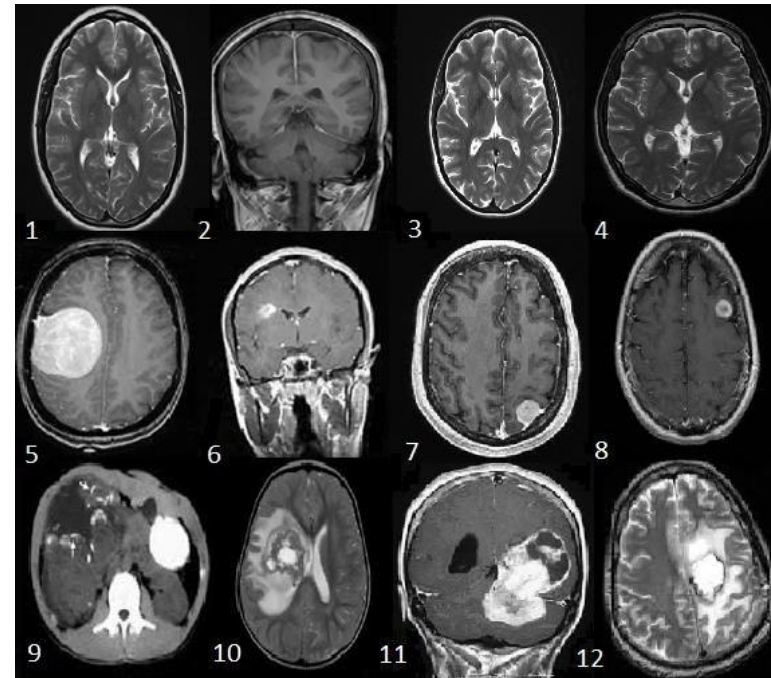


Electromagnetic Spectrum

Type of Radiation	Frequency Range (Hz)	Wavelength Range
Gamma-rays	$10^{20} - 10^{24}$	$< 10^{-12}$ m
X-rays	$10^{17} - 10^{20}$	1 pm – 1 nm
Ultraviolet	$10^{15} - 10^{17}$	1 nm – 400 nm
Visible	$4 \times 10^{14} - 7.5 \times 10^{14}$	400 nm – 750 nm
Near-infrared	$1 \times 10^{14} - 4 \times 10^{14}$	750 nm – 2.5 μ m
Infrared	$10^{13} - 10^{14}$	2.5 μ m – 25 μ m
Microwaves	$3 \times 10^{11} - 10^{13}$	25 μ m – 1 mm
Radio waves	$< 3 \times 10^{11}$	> 1 mm

Radio Waves Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
Radio waves	$< 3 \times 10^{11}$	$> 1 \text{ mm}$	AM/FM radio, Computer Network for transmission of data. In Image Processing: Remote sensing and medical imaging (MRI)



Microwaves Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
Microwave	$3 \times 10^{11} - 10^{13}$	$25 \mu\text{m} - 1 \text{ mm}$	Radar imaging, weather forecasting, imaging planetary surfaces. Medical Imaging: to probe the magnetic resonance (MR) properties of electrons.



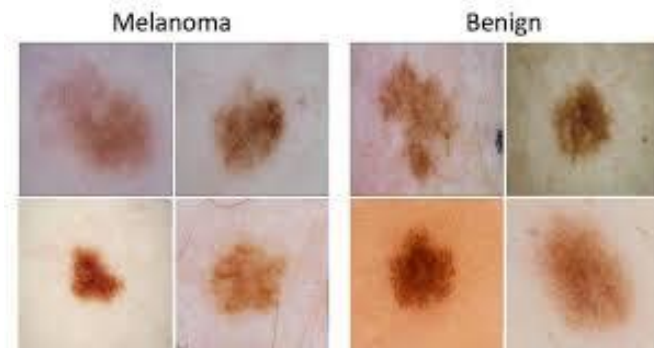
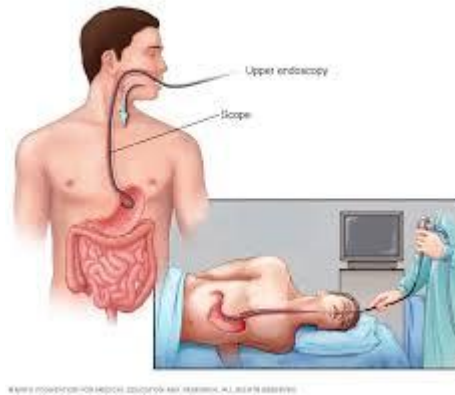
Infrared Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
Near-infrared	$1 \times 10^{14} - 4 \times 10^{14}$	750 nm – 2.5 μm	Thermal imaging, Night vision technology. Medical imaging: cancer stage detection.
Infrared	$10^{13} - 10^{14}$	2.5 $\mu\text{m} - 25 \mu\text{m}$	



Visible Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
Visible Light	4×10^{14} – 7.5×10^{14}	400 nm – 750 nm	Optical , Photography. Medical Imaging: Endoscopy and dermoscopy.



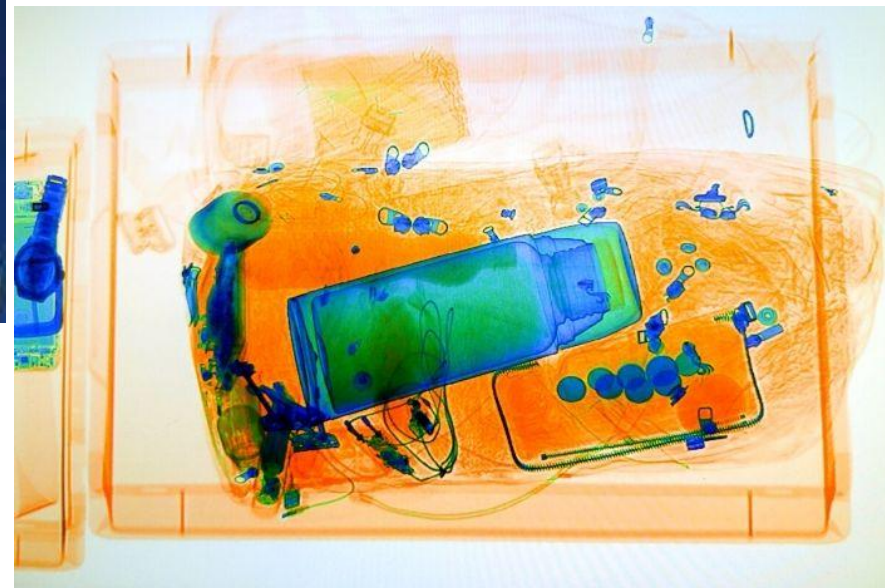
Ultraviolet Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
Ultraviolet	$10^{15} - 10^{17}$	1 nm – 400 nm	Optical , lithography, microscopy laser



X-rays Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
X-rays	$10^{17} - 10^{20}$	1 pm – 1 nm	Medical, analysis of electrophoresis gels, Chromosome analysis, Bacterial colony counting and autoradiography analysis



Gamma-Rays Imaging

Types of Radiation	Frequency Range (Hz)	Wavelength	Nature of imaging and its relevance for image processing
Gamma rays	$10^{20} - 10^{24}$	$< 10^{-12}$ m	Medical (Computed Tomography), and in industry (for finding tiny cracks in metal surface)

Computed Tomography (CT)



Gamma-Rays Imaging

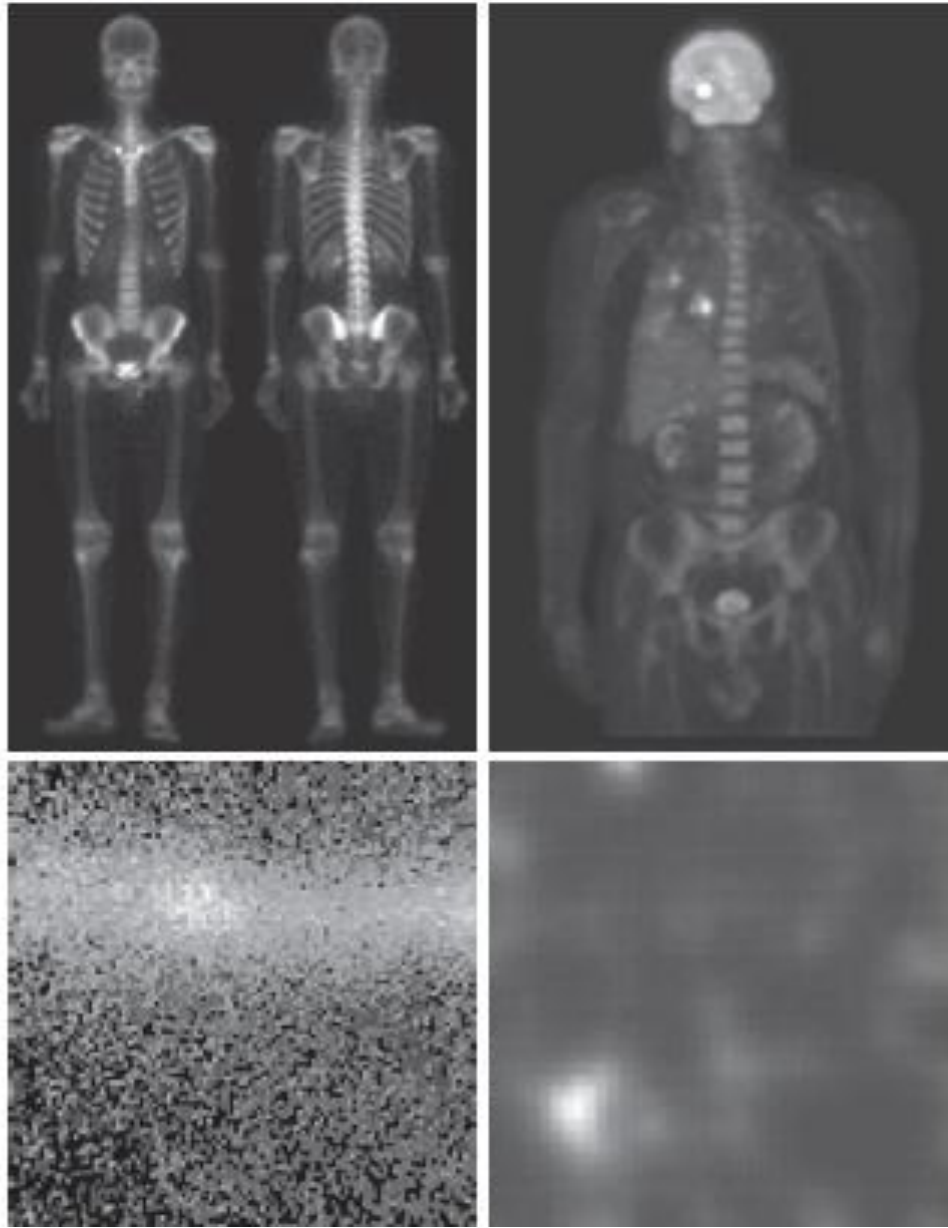
a b
c d

FIGURE 1.6

Examples of gamma-ray imaging.

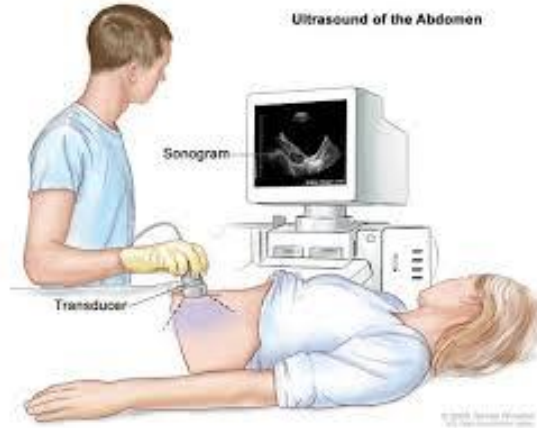
- (a) Bone scan.
- (b) PET image.
- (c) Cygnus Loop.
- (d) Gamma radiation (bright spot) from a reactor valve.

(Images courtesy of (a) G.E. Medical Systems; (b) Dr. Michael E. Casey, CTI PET Systems; (c) NASA; (d) Professors Zhong He and David K. Wehe, University of Michigan.)



Ultrasound Imaging

Sound is a longitudinal wave, and it is possible to detect objects and produce images with sound.





Thank You